MANUFACTURING AND PLANT

Case Study Options

ELIGIBLE DISCIPLINES

Open to pre-final and final-year students with Bachelor's degree in relevant disciplines, including Mechanical, Chemical, Electrical, Food Tech, Metallurgy & Production.



OPTION 01

Safer Work, Better Work

CONTEXT

Manufacturing work involves equipment, materials, movement of people and goods, and routine tasks. A plant wants to make everyday work safer while keeping operations practical and efficient.

CHALLENGE

Identify a workplace risk and propose an improvement that helps prevent incidents and supports a strong safety culture.

CONSIDER

1. How can employees spot and report risks easily?
2. Could equipment design, training, visual controls, or digital tools help?
3. How will you ensure the solution works in daily practice?
4. How could the safety improvement also reduce energy use, material waste, or other environmental impacts?

DELIVERABLES

Risk analysis, proposed improvement, implementation plan, and measures such as near-miss reporting, task-risk reduction, and employee feedback.

OPTION 02

Getting Quality Right First Time

CONTEXT

Customers expect products to be consistent and reliable. Variations in materials, equipment, processes, or human practices can lead to defects, rework, or delays.

CHALLENGE

Develop a way to prevent a recurring quality problem earlier in the process rather than relying only on final checks.

CONSIDER

1. Where could variation enter the process?
2. How can employees detect issues sooner?
3. How can data help identify patterns and root causes?
4. How could preventing defects and rework reduce the use of materials, water, energy, or packaging?

DELIVERABLES

Problem definition, root-cause analysis, prevention proposal, and measures such as defects, rework, and first-pass quality.

OPTION 03

Less Waste, Smarter Resource Use

CONTEXT

Manufacturing uses materials, water, energy, and packaging. Small losses repeated across many production runs can add up, while changes to reduce waste must still protect quality and safety.

CHALLENGE

Choose one source of waste or resource use and design a realistic improvement.

CONSIDER

1. Where does the loss occur?
2. Can process, equipment, planning, or employee practices reduce it?
3. What trade-offs should be checked?

DELIVERABLES

Baseline data needed, proposed solution, pilot plan, and measures such as material loss, energy or water use, and waste generated.